Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Lecture 1: Dominance and Iterated Elimination (graduate level)

Answer Key and Rubric

Instructions. Multiple-choice items: mark the single correct option (no penalty for errors). *True/False* items: every 2 wrong items cancel 1 correct one (score = $\max\{0, \text{correct} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step; a result the item asks you to derive may not be quoted. All questions are symbolic: state the admissible range of every parameter you use. Throughout, "dominated" means dominated by another *pure* strategy, as in Lecture 1; "IESDS" removes, at every round, every strictly dominated strategy of every player. Lecture 2 material (Nash equilibrium, mixed strategies) is neither needed nor allowed.

Block 1 — Normal form, dominance and social dilemmas *(Lecture 1 • 25 points)*

Question 1. (10 points) Objective items.

a) (3 points) *Dominance × Pareto optimality.* A medical clinic in Recife is owned by $n \geq 2$ partners. Each partner simultaneously either exerts effort (E, private cost $c > 0$) or shirks (S). If k partners exert effort, the clinic's net revenue is $Y(k) = yk + \theta\binom{k}{2}$, with $y > 0$ and $\theta \geq 0$ (every pair of active partners adds θ through cross-referrals), and revenue is split equally. The set of costs c for which shirking is strictly dominant for every partner *and* the resulting profile is Pareto dominated by some other profile is:

A. $(y/n,\ y)$

B. $\big((y + \theta(n-1))/n,\ y + \theta(n-1)/2\big)$

C. $\big((y + \theta(n-1))/n,\ y + \theta(n-1)\big)$

D. $(y/n,\ y + \theta(n-1)/2)$

E. $\big((y + \theta)/n,\ y + \theta n/2\big)$

Answer: B

Step — marginal return. Fix the number $k \in \{0, \ldots, n-1\}$ of *other* partners who exert effort. Switching from S to E raises revenue by $Y(k+1) - Y(k) = y + \theta k$, of which the partner keeps $(y + \theta k)/n$, at cost c.

Step — dominance. S strictly dominates E iff $(y + \theta k)/n < c$ for *every* $k \in \{0, \ldots, n-1\}$; since $\theta \geq 0$ the binding case is $k = n - 1$: $c > (y + \theta(n-1))/n$. With two strategies, S is then strictly dominant and all-S is the unique strict dominant-strategy equilibrium, with payoff 0 for everyone.

Step — Pareto. With $k \geq 1$ active partners, the active ones get $Y(k)/n - c$ and the idle ones $Y(k)/n$. Y is strictly increasing, so $Y(k)/n - c \leq Y(n)/n - c$ with equality only at $k = n$. If $c < Y(n)/n$, all-E gives everyone $Y(n)/n - c > 0$: it Pareto dominates all-S. If $c \geq Y(n)/n$, every profile with $k \geq 1$ gives each active partner $Y(k)/n - c < 0$ for $k < n$, and everyone ≤ 0 at $k = n$: no profile Pareto dominates all-S. Hence the condition is $c < Y(n)/n = y + \theta(n-1)/2$.

Step — non-empty. $y + \frac{\theta(n-1)}{2} - \frac{y+\theta(n-1)}{n} = \frac{(n-1)y}{n} + \frac{\theta(n-1)(n-2)}{2n} > 0$ for $n \geq 2$.

Step — distractors. A sets $\theta = 0$ in both bounds (the lecture's partnership); D checks dominance only against $k = 0$, where effort pays least (necessary, not sufficient); C compares with the last partner's marginal revenue instead of per-capita revenue; E counts θ once per partner instead of once per pair, in both bounds.

Economic intuition. Synergy makes effort most valuable when everyone else works, so dominance must hold in that best case; the dilemma needs full cooperation to pay per capita.

Ref: Lecture 1, "Examples" (partnership); Tadelis (2013) §4.1.2–4.1.3, pp. 61–62, and §3.3.3, p. 56.

b) (3 points) *Weak dominance × IESDS.* A court-ordered sale (*leilão judicial*) of a warehouse in Campinas uses a sealed-bid second-price rule with a public reserve price $r > 0$: $n \geq 2$ bidders submit bids $b_j \in [0, \infty)$; if every bid is below r there is no sale; otherwise the highest bidder wins (ties broken uniformly at random) and pays $\max\{r, \text{highest other bid}\}$. Bidder i values the warehouse at $v_i > r$. Which statement is correct?

A. Bidding v_i strictly dominates every other bid, so IESDS selects truthful bidding for every bidder whose value exceeds r.

B. Bidding v_i weakly dominates the bids above v_i only; bids in $[r, v_i)$ are not weakly dominated, since they lower the price paid.

C. Bidding $v_i - r$ weakly dominates every other bid, because the reserve price is a cost that the winner must net out of the value.

D. Bidding v_i weakly dominates every other bid, yet no bid of any bidder is strictly dominated, so IESDS removes no strategy at all.

E. Bidding v_i weakly dominates every other bid, and the bids below r are strictly dominated by v_i, so IESDS removes them.

Answer: D

Step — price is not one's own bid. Let $p \equiv \max\{r, h\}$, h the highest rival bid. A winning bidder pays p, which does not depend on her own bid; her bid only decides *whether* she wins.

Step — weak dominance. If $p < v_i$: bidding v_i wins and earns $v_i - p > 0$; any $b > p$ earns the same, $b < p$ earns 0, $b = p$ earns at most $v_i - p$. If $p > v_i$: v_i loses (0); any winning bid pays $p > v_i$ (negative). If $p = v_i$: every bid earns 0. So v_i is never worse, and it is strictly better against rival bids with $p \in (b, v_i)$ (for $b < v_i$) or with $h \in (v_i, b)$ (for $b > v_i$).

Step — no strict dominance. Take any two bids $b \neq b'$ of any bidder j and the rival profile in which someone bids above $\max\{b, b'\}$: both earn 0. No inequality is strict for all rival profiles, so no bid is strictly dominated: $S_j^1 = S_j$ and IESDS has no bite.

Step — distractors. A confuses weak with strict dominance; B assumes the bid sets the price; C subtracts the reserve from the value; E calls the bids below r strictly dominated, but they tie with v_i whenever a rival bids above v_i.

Economic intuition. Truthful bidding is robust (never a mistake), but the argument behind it uses weak dominance; common knowledge of rationality alone, through IESDS, says nothing in this auction.

Ref: Tadelis (2013) §13.1.1, pp. 272–273 (Proposition 13.1, no reserve); Lecture 1, "Examples" (second-price auction) and "Dominance".

c) (2 points, T or F) *Invariance × dominance.* Let G be a finite game with a strict dominant-strategy equilibrium s^D. Then there is $\varepsilon > 0$ such that s^D is a strict dominant-strategy equilibrium of every game G' with the same players and strategy sets whose payoffs satisfy $|u_i'(s) - u_i(s)| < \varepsilon$ for every i and every s.

Answer: T

Step — proof. Let $g \equiv \min\{u_i(s_i^D, s_{-i}) - u_i(s_i, s_{-i}) : \ i \in I, \ s_i \neq s_i^D, \ s_{-i} \in S_{-i}\}$, a minimum over a finite set of strictly positive numbers, so $g > 0$. Take $\varepsilon = g/2$. Then $u_i'(s_i^D, s_{-i}) - u_i'(s_i, s_{-i}) > g - 2\varepsilon = 0$ for every i, $s_i \neq s_i^D$, s_{-i}.

Step — what breaks it. Finiteness: with infinite strategy sets the infimum gap can be 0 (in Question 2 the gap is $\frac{c}{2}(e_i - e_i^D)^2$). Strictness: if U weakly dominates D with $u_1(U, L) = u_1(D, L)$, raising $u_1(D, L)$ by $\varepsilon/2$ destroys the weak dominance for every $\varepsilon > 0$.

Ref: Tadelis (2013) §3.3.2, pp. 55–56 (invariance); §4.1.3, p. 62.

d) (2 points, T or F) *Pareto optimality × dominance.* In every finite game, with any number of players, that has a strict dominant-strategy equilibrium s^D, every profile that Pareto dominates s^D has *every* player i using a strategy different from s_i^D.

Answer: F

Step — true for two players. If s' Pareto dominates s^D and $s_1' = s_1^D$, then $s_2' \neq s_2^D$ and $u_2(s') = u_2(s_1^D, s_2') < u_2(s^D)$ by strict dominance: contradiction. Symmetrically for player 2.

Step — counterexample with three players. Players 1 and 2 play a prisoner's dilemma whose payoffs do not depend on player 3: $(M, M) \mapsto (-1, -1)$, $(M, F) \mapsto (-4, 0)$, $(F, M) \mapsto (0, -4)$,

$(F,F) \mapsto (-3,-3)$. Player 3 chooses X or Y with $u_3 = u_1 + u_2 + \mathbb{1}\{s_3 = X\}$, so X is strictly dominant and $s^D = (F,F,X)$. The profile (M,M,X) gives $(-1,-1,-1)$ against $(-3,-3,-5)$: it Pareto dominates s^D while player 3 keeps X. With $n \geq 3$ a player who does not deviate can gain from the others' deviations.

Ref: Lecture 1, "Dominance and social dilemmas"; Tadelis (2013) §4.1.3, p. 62.

Question 2. (15 points) *Dominance × Pareto optimality and social dilemmas × solution-concept criteria.* An accounting firm in Belo Horizonte is a partnership of $n \geq 2$ partners. Simultaneously, each partner i chooses an effort level $e_i \geq 0$. Firm revenue is $y \sum_j e_j$ $(y > 0)$; partner i bears the private cost $\frac{c}{2}e_i^2$ $(c > 0)$ and receives the share $\alpha_i \geq 0$ of revenue, with $\sum_i \alpha_i = 1$ fixed in the partnership agreement. Efforts are not observable by the other partners, so nothing else can be contracted on; each partner cares only about her own payoff. Unless stated otherwise, $\alpha_i = 1/n$.

a) (3 points) Write the game in normal form $G = (I, \{S_i\}, \{u_i\})$. Prove that every partner has a strictly dominant strategy, for any shares (α_i), and find it. Draw, in the (e_i, u_i) plane, partner i's payoff as a function of her own effort for two different values of the others' total effort, and read the dominance property off the figure.

Step — normal form. $I = \{1,\dots,n\}$, $S_i = [0,\infty)$, and

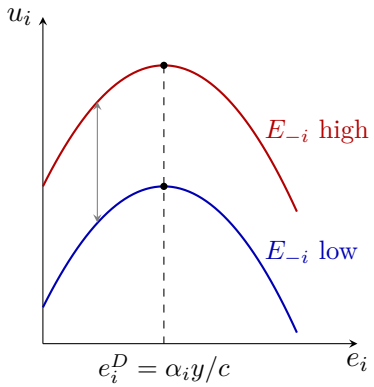
$$u_i(e) = \alpha_i y \sum_j e_j - \tfrac{c}{2}e_i^2 = \underbrace{\alpha_i y \textstyle\sum_{j\neq i} e_j}_{\text{others}} + \alpha_i y e_i - \tfrac{c}{2}e_i^2.$$

Step — dominance. Let $e_i^D \equiv \alpha_i y / c$. For any e_i and any e_{-i},

$$u_i(e_i^D, e_{-i}) - u_i(e_i, e_{-i}) = \alpha_i y (e_i^D - e_i) - \tfrac{c}{2}\left((e_i^D)^2 - e_i^2\right) = \tfrac{c}{2}\,(e_i - e_i^D)^2,$$

using $\alpha_i y = c e_i^D$. The difference does not depend on e_{-i} and is strictly positive for $e_i \neq e_i^D$: e_i^D is strictly dominant (with $\alpha_i = 0$ it is $e_i^D = 0$).

Step — graph. The others' effort enters additively, so a change in $E_{-i} = \sum_{j\neq i} e_j$ shifts the whole parabola vertically by $\alpha_i y\, \Delta E_{-i}$ (gray arrow) without moving its peak.



Step — reading. Whatever partner i believes about the others, her payoff curve peaks at the same e_i^D: the ranking of her own strategies is belief-free, which is what dominance means.

Ref: Tadelis (2013) §3.1, pp. 46–49; §4.1.1–4.1.2, pp. 60–61.

Grading rubric.

- Normal form with S_i and u_i: 0.5 pt.
- Dominance proved for *every* e_{-i} (difference identity or equivalent): 1.25 pts. Only the first-order condition $\alpha_i y = c e_i$ (best reply, not dominance): max. 1 pt on the item.
- Graph: labelled axes 0.25, two parabolas with the same peak 0.5, vertical shift labelled 0.25; reading 0.25. A graph contradicting the algebra loses the graph points.

b) (3 points) Take equal shares. Show that the game has a unique strict dominant-strategy equilibrium and a unique iterated-elimination equilibrium. Prove that this profile is Pareto dominated, and find the set of symmetric effort levels $\bar{e}$ such that $(\bar{e}, \dots, \bar{e})$ Pareto dominates it. Which symmetric profile is best for every partner?

Step — equilibrium. From a), $e^D = y/(nc)$ for everyone. A strict dominant-strategy equilibrium is unique (Lecture 1 proposition), and in round 1 of IESDS every $e_i \neq e^D$ is strictly dominated by e^D, so $S_i^1 = \{e^D\}$: the unique iterated-elimination equilibrium.

Step — symmetric payoffs. In $(\bar{e}, \dots, \bar{e})$ each partner receives $\frac{1}{n}\, y\, n\bar{e}$ and pays $\frac{c}{2}\bar{e}^2$: $u(\bar{e}) = y\bar{e} - \frac{c}{2}\bar{e}^2$. At e^D: $u^D = \dfrac{y^2}{nc} - \dfrac{y^2}{2n^2c} = \dfrac{y^2(2n-1)}{2n^2c}$.

Step — dominating set. $u(\bar{e}) > u^D \iff \frac{c}{2}\bar{e}^2 - y\bar{e} + u^D < 0$. The discriminant is $y^2 - 2cu^D = y^2(n-1)^2/n^2 > 0$ for $n \geq 2$, with roots $\frac{y}{c}(1 \pm \frac{n-1}{n})$:

$$\bar{e} \in \Big(\frac{y}{nc},\ \frac{(2n-1)\,y}{nc}\Big).$$

All partners gain in the same proportion, so these are exactly the symmetric Pareto improvements.

Step — best symmetric profile. $u(\bar{e})$ is maximized at $\bar{e} = y/c \equiv e^*$, the midpoint of the interval, with $u = y^2/(2c)$. It is Pareto optimal: $\sum_i u_i = y\sum_j e_j - \frac{c}{2}\sum_j e_j^2$ (shares sum to 1) is maximized at $e_j = y/c$ for all j, and any Pareto improvement would raise the sum.

Economic intuition. Each partner keeps $1/n$ of the value of her effort but bears all of its cost: individually dominant, collectively dominated. This is the prisoner's dilemma with a continuum of strategies.

Ref: Lecture 1, "Dominance and social dilemmas"; Tadelis (2013) §4.1.3, p. 62; M&R Ch. 2, "Efficiency", pp. 22–24.

Grading rubric.

- e^D and both uniqueness claims (strict DSE and round-1 IESDS): 0.75 pt.
- u^D: 0.5 pt. Dominating interval with both roots: 1 pt.
- $e^* = y/c$ and its Pareto optimality argued: 0.75 pt.
- Only e^* shown to dominate (no interval): max. 2 pts. Arithmetic slip with right method: -0.5 pt.

c) (3 points) Return to general shares. Prove that no sharing rule ($\alpha_i \geq 0$, $\sum_i \alpha_i = 1$) makes the efficient profile $(y/c, \dots, y/c)$ a dominant-strategy equilibrium. Characterize the sharing rule that maximizes total surplus $W \equiv \sum_i u_i$ at the dominant-strategy equilibrium, and the rule that minimizes it.

Step — impossibility. From a), $e_i^D = \alpha_i y/c = y/c \iff \alpha_i = 1$. Budget balance allows this for at most one partner, and then $\alpha_j = 0$ and $e_j^D = 0 \neq y/c$ for the other $n-1 \geq 1$ partners.

Step — surplus at the DSE. With $e_j^D = \alpha_j y/c$: $W(\alpha) = y\sum_j e_j^D - \frac{c}{2}\sum_j (e_j^D)^2 = \dfrac{y^2}{c}\sum_j (\alpha_j - \frac{1}{2}\alpha_j^2)$.

Step — maximization. $\max W$ s.t. $\sum_j \alpha_j = 1$, $\alpha_j \geq 0$. The objective is a sum of strictly concave functions. Lagrangian $\sum_j(\alpha_j - \frac{1}{2}\alpha_j^2) - \mu(\sum_j \alpha_j - 1)$; FOC $1 - \alpha_j = \mu$ for every j with $\alpha_j > 0$, so all positive shares are equal. A zero share would need $1 - 0 \leq \mu = 1 - \frac{1}{n'} < 1$ (n' the number of positive shares): impossible. Hence the unique maximizer is $\alpha_j = 1/n$, with $W = \dfrac{y^2}{c}\Big(1 - \dfrac{1}{2n}\Big)$.

Step — minimization. A concave function on the simplex attains its minimum at a vertex: $\alpha = (1, 0, \dots, 0)$, one partner works at y/c and the rest at 0, $W = y^2/(2c)$.

Economic intuition. Equal split is the best a budget-balanced partnership can do, and it is still inefficient: the inefficiency comes from budget balance itself, since every partner would need the whole marginal product, $\sum_i \alpha_i = n$, which only an outside party could fund.

Ref: Tadelis (2013) §3.3.3, p. 56; M&R Ch. 2, "Efficiency", pp. 22–24.

Grading rubric.

- Impossibility with the budget-balance argument: 1 pt.
- $W(\alpha)$: 0.5 pt. Equal split proved optimal (concavity + FOC, corners excluded): 1 pt.
- Minimum at a vertex: 0.5 pt.
- Equal split asserted "by symmetry" without proof: 0 for that line.

d) (3 points) With equal shares, quantify the social dilemma: total surplus at the efficient profile, W^*, and at the dominant-strategy equilibrium, W^D; the relative loss L/W^*, where $L \equiv W^* - W^D$; the payoff gap per partner; and the effort ratio e^D/e^*.

Step — levels. $W^* = n\,u(e^*) = \dfrac{ny^2}{2c}$; $W^D = n\,u^D = \dfrac{y^2(2n-1)}{2nc}$.

Step — loss. $\dfrac{L}{W^*} = 1 - \dfrac{2n-1}{n^2} = \dfrac{(n-1)^2}{n^2}$; per partner, $u(e^*) - u^D = \dfrac{y^2(n-1)^2}{2n^2c}$; effort ratio $e^D/e^* = 1/n$.

Step — benchmark. $n = 2$: a quarter of the surplus is lost; $n = 5$: 64%.

Economic intuition. Effort falls to $1/n$ of its efficient level, and the surplus loss is quadratic in the shortfall: the dilemma deepens fast with the number of partners.

Ref: Lecture 1, "Dominance and social dilemmas"; Tadelis (2013) §3.3.3, p. 56.

Grading rubric.

- W^* and W^D: 1.25 pts. L/W^* simplified: 1 pt. Gap and ratio: 0.75 pt.
- W^D computed with the efficient effort for one partner and e^D for the others: max. 1 pt.

e) (3 points) Parameter analysis. Prove the sign of the effect of y, c and n (treated as continuous) on e^D, e^*, the per-partner payoff u^D, total surplus W^D and the relative loss L/W^*. Summarize in a table, study $n \to 1$, $n \to \infty$ and $c \to 0$, and give the economic reading.

Step — signs. $e^D = y/(nc)$: $+y$, $-c$, $-n$. $e^* = y/c$: $+y$, $-c$, no n. $u^D = \frac{y^2}{c}\frac{2n-1}{2n^2}$: $\partial_n u^D = -\dfrac{y^2(n-1)}{n^3c} < 0$ for $n > 1$. $W^D = \frac{y^2}{c}(1 - \frac{1}{2n})$: $\partial_n W^D = \dfrac{y^2}{2n^2c} > 0$. $L/W^* = (1-1/n)^2$: $\partial_n = 2(n-1)/n^3 > 0$; no y, no c.

	y	c	n
e^D	$+$	$-$	$-$
e^*	$+$	$-$	0
u^D	$+$	$-$	$-$
W^D	$+$	$-$	$+$
L/W^*	0	0	$+$

Step — limits. $n \to 1$: $e^D \to e^*$, $L \to 0$ (a sole proprietor faces no dilemma). $n \to \infty$: $e^D, u^D \to 0$, $W^D \to y^2/c$ (bounded) while $W^* \to \infty$, so $L/W^* \to 1$. $c \to 0$: every level diverges, L/W^* is unchanged: the loss share is a pure function of n.

Step — reading. Productivity and cost scale the stakes but not the distortion; only the number of claimants on each unit of output does. More partners raise total effort (W^D rises) but lower each one's effort and payoff. The prediction is invariant in the sense of Lecture 1: small changes in y or c move e^D continuously, and the dominance argument never uses beliefs.

Grading rubric.

- Signs proved: 1.5 pts (the W^D-in-n sign, opposite to u^D, counts 0.5).
- Table 0.25 pt; limits 0.5 pt; reading 0.75 pt.
- Signs listed without proof or reading: max. half the item.

Block 2 — Iterated elimination of strictly dominated strategies *(Lecture 1 • 25 points)*

Question 3. (10 points) Objective items.

a) (3 points) *IESDS × common knowledge of rationality.* A distributor (player 1, rows) and a retailer (player 2, columns) in the Zona Franca de Manaus play the game below, in which $x \in \mathbb{R}$ is the distributor's payoff at (U, C):

	L	C	R
U	$2, 3$	$x, 2$	$5, 0$
M	$5, 3$	$4, 4$	$2, 2$
D	$1, 4$	$1, 5$	$4, 6$

IESDS selects a unique profile if and only if, and the number of rounds it takes is:

A. $x > 1$; four rounds

B. $x > 1$; three rounds

C. $1 < x < 4$; three rounds

D. $x < 4$; two rounds

E. $1 < x < 4$; four rounds

Answer: E

Step — round 1. U vs D column by column: $2 > 1$, x against 1, $5 > 4$; so U strictly dominates D iff $x > 1$. M does not dominate D ($2 < 4$ at R). Player 2: no column dominates another (L is best against U, R against D, and C beats L at M). If $x \leq 1$ nothing is removed and $S^{\infty} = S$.

Step — round 2 ($x > 1$). Rows $\{U, M\}$: R gives $(0, 2)$, strictly below L $(3, 3)$: R removed. U vs M still compares column R ($5 > 2$): nothing removed for player 1.

Step — round 3. Columns $\{L, C\}$: M $(5, 4)$ strictly dominates U $(2, x)$ iff $x < 4$.

Step — round 4. Row M: C $(4) > L$ (3); the survivor is (M, C). If $x \geq 4$ the process stops at $\{U, M\} \times \{L, C\}$. So: unique iff $1 < x < 4$, after four rounds; the last round uses third-order knowledge of rationality.

Step — distractors. A and B drop the condition $x < 4$; C and D miscount rounds (C also merges rounds 2 and 3, which are sequential because U vs M hinges on column R); D also drops $x > 1$.

Ref: Tadelis (2013) §4.2.1, pp. 63–65; Gibbons (1992) §1.1.B, pp. 4–8.

b) (3 points) *Weak dominance × order of elimination.* In the game below, consider iterated elimination of *weakly* dominated strategies, removing one strategy at a time in any order until no strategy is weakly dominated:

	L	C	R
U	$1, 2$	$1, 0$	$0, 1$
M	$1, 2$	$2, 2$	$2, 1$
D	$0, 0$	$2, 1$	$1, 2$

Which statement is correct?

A. Some orders end at the single profile (M, L) and others at the single profile (M, C), so player 1's final payoff (1 or 2) depends on the order.

B. Every order ends at (M, C), because M weakly dominates both U and D, after which player 2 prefers C.

C. IESDS (strict) already selects (M, C), and every order of weak elimination agrees with it.

D. Removing all weakly dominated strategies at once in every round ends at the single profile (M, C).

E. Some orders end at (M, L) and others at (U, L), so player 1's final payoff is 1 in every order.

Answer: A

Step — strict. No strategy is strictly dominated (U ties M at L, D ties M at C; no column is dominated), so IESDS removes nothing: C is false.

Step — weak, first step. M weakly dominates U (tie at L) and D (tie at C); no column is weakly dominated in the full game.

Step — order 1. Remove U; on rows $\{M, D\}$, C $(2,1)$ weakly dominates L $(2,0)$: remove L; on columns $\{C, R\}$, M $(2,2)$ weakly dominates D $(2,1)$: remove D; then R $(1) < C$ (2): end at (M, C), payoffs $(2,2)$.

Step — order 2. Remove D; on rows $\{U, M\}$, L $(2,2)$ weakly dominates C $(0,2)$: remove C; on columns $\{L, R\}$, M $(1,2)$ weakly dominates U $(1,0)$: remove U; then R $(1) < L$ (2): end at (M, L), payoffs $(1,2)$.

Step — other claims. All at once: round 1 removes U and D; then R; L and C tie for player 2: $\{M\} \times \{L, C\}$, not a singleton (D false). M is never removed, and a run that keeps U ends at $\{U, M\} \times \{L\}$, never at the single profile (U, L) (E false); B is refuted by order 2.

Economic intuition. A weakly dominated strategy can be the only reason another strategy is weakly dominated; removing it first changes what is removed later. Strict dominance does not have this problem (item d).

Ref: Tadelis (2013) §4.1.3, pp. 62–63; Lecture 1, "Dominance" (weak dominance).

c) (2 points, T or F) *IESDS × existence.* In every normal-form game, IESDS leaves every player at least one strategy: $S_i^\infty \neq \varnothing$ for every i.

Answer: F

Step — counterexample. Let $S_1 = [0, 1)$ and $u_1(s) = s_1$ (player 2 arbitrary). Every s_1 is strictly dominated by $(1 + s_1)/2 \in (s_1, 1)$, so $S_1^1 = \varnothing$ already in round 1.

Step — finite games. With finite sets, the strategy $\hat{s}_i$ that maximizes $\sum_{s_{-i} \in S_{-i}^{k-1}} u_i(s_i, s_{-i})$ over S_i^{k-1} cannot be strictly dominated within S^{k-1} (a dominator would have a strictly larger sum), so $S_i^k \neq \varnothing$, and the process stops after finitely many rounds. The existence claim of the lecture is a claim about finite games.

Ref: Tadelis (2013) §4.2.3, pp. 67–68; §3.3.2, p. 55 (existence).

d) (2 points, T or F) *IESDS × order of elimination.* In a finite game, removing at each step *some* (not necessarily all) of the strictly dominated strategies, in any order, until none is left always ends at the same sets S_i^∞ as the lecture's procedure.

Answer: T

Step — setup. Let A be the lecture's result and B the end of any other run; both are terminal (no strictly dominated strategy relative to themselves). Show $B \subseteq A$; $A \subseteq B$ is the same argument with the roles swapped.

Step — proof. Suppose some strategy of B is removed by the lecture's procedure; let $s_i \in B_i$ be one removed first, at a round whose current sets contain B. It was dominated by some s_i' relative to a superset of B_{-i}, so the set $\mathcal{D} = \{t \in S_i : \ t \text{ strictly dominates } s_i \text{ on } B_{-i}\}$ is non-empty. Pick $t^* \in \mathcal{D}$ maximizing $\sum_{s_{-i} \in B_{-i}} u_i(t, s_{-i})$ (finite). If $t^* \in B_i$, s_i is dominated within B: contradiction. If $t^* \notin B_i$, the other run removed it, dominated by some t' relative to a superset of B_{-i}; then t' dominates s_i on B_{-i}, so $t' \in \mathcal{D}$ with a larger sum: contradiction.

Ref: Tadelis (2013) §4.2.1, pp. 63–65; Gibbons (1992) §1.1.B, pp. 4–8.

Question 4. (15 points) *IESDS × common knowledge of rationality × solution-concept criteria (uniqueness, invariance).* Two cement producers supply a regional market in the North of Brazil and choose quantities simultaneously. Inverse demand is $P(Q) = \max\{a - bQ, 0\}$ with $Q = q_1 + q_2$; firm i has constant marginal cost $c_i \in (0, a)$ and no fixed cost; $S_i = [0, \infty)$ and $u_i = \pi_i(q_i, q_j) = q_i(P(q_i + q_j) - c_i)$.

a) (3 points) Show that, for each $q_j \geq 0$, $\pi_i(\cdot, q_j)$ is strictly increasing up to a unique profit-maximizing quantity $\mathrm{BR}_i(q_j)$ and strictly decreasing beyond it, and find BR_i. Determine S_i^1 exactly, and say where the assumption $c_i > 0$ is used.

Step — two regions. If $q_i + q_j < a/b$: $\pi_i = q_i(a - c_i - bq_j) - bq_i^2$, with $\partial\pi_i/\partial q_i = a - c_i - bq_j - 2bq_i$, strictly decreasing in q_i. If $q_i + q_j \geq a/b$: $\pi_i = -c_i q_i$, strictly decreasing because $c_i > 0$. π_i is continuous at the kink.

Step — best reply. Hence $\pi_i(\cdot, q_j)$ rises up to $\mathrm{BR}_i(q_j) = \max\{\frac{a-c_i-bq_j}{2b}, 0\}$ and falls strictly afterwards (the kink lies beyond $\mathrm{BR}_i(q_j)$ since the price there is $0 < c_i$). BR_i is non-increasing with slope $-\frac{1}{2}$ or 0.

Step — round 1. $\mathrm{BR}_i(q_j) \leq \mathrm{BR}_i(0) = m_i \equiv \frac{a-c_i}{2b}$. For $q > m_i$ and every q_j, both m_i and q lie in the decreasing part with $m_i < q$: $\pi_i(m_i, q_j) > \pi_i(q, q_j)$, so q is strictly dominated by m_i. Each $q \in [0, m_i]$ is the *unique* maximizer against $q_j = (a - c_i - 2bq)/b \geq 0$, so it cannot be strictly dominated. $S_i^1 = [0, m_i]$.

Step — role of $c_i > 0$. With $c_i = 0$, every q_i earns 0 against $q_j \geq a/b$, so $q > m_i$ is only weakly dominated and round 1 removes nothing for firm i; elimination starts a round later.

Ref: Tadelis (2013) §4.2.2, pp. 65–67; Gibbons (1992) §1.2.A, pp. 18–20.

Grading rubric.

- Both regions and single-peakedness: 1 pt (the zero-price region ignored: max. 2 pts on the item).
- BR_i with the $\max\{\cdot, 0\}$: 0.5 pt.
- $S_i^1 = [0, m_i]$ with both directions (dominated above, undominated below): 1 pt.
- Role of $c_i > 0$: 0.5 pt.

b) (3 points) Show that every S_i^k is a closed interval $[L_i^k, U_i^k]$ and derive the recursion linking its end points to those of S_j^{k-1}. Interpret round k in terms of knowledge of rationality.

Step — claim. By induction, $S_i^k = [\mathrm{BR}_i(U_j^{k-1}), \mathrm{BR}_i(L_j^{k-1})]$, with $U_j^0 = \infty$, $L_j^0 = 0$ and $\mathrm{BR}_i(\infty) \equiv 0$ (this gives $S_i^1 = [0, m_i]$). The new end points lie in S_i^{k-1} because BR_i is non-increasing and the intervals are nested.

Step — below the lower end. Let $q < \mathrm{BR}_i(U_j^{k-1})$. For every $q_j \in S_j^{k-1}$, $\mathrm{BR}_i(q_j) \geq \mathrm{BR}_i(U_j^{k-1}) > q$, and $\pi_i(\cdot, q_j)$ is strictly increasing on $[0, \mathrm{BR}_i(q_j)]$: q is strictly dominated by $\mathrm{BR}_i(U_j^{k-1})$.

Step — above the upper end. Let $q > \mathrm{BR}_i(L_j^{k-1})$. For every $q_j \in S_j^{k-1}$, $\mathrm{BR}_i(q_j) \leq \mathrm{BR}_i(L_j^{k-1}) < q$, and π_i is strictly decreasing beyond $\mathrm{BR}_i(q_j)$: q is strictly dominated by $\mathrm{BR}_i(L_j^{k-1})$.

Step — in between. BR_i is continuous; by the intermediate value theorem each

$$q \in [\mathrm{BR}_i(U_j^{k-1}),\ \mathrm{BR}_i(L_j^{k-1})]$$

equals $\mathrm{BR}_i(q_j)$ for some $q_j \in S_j^{k-1}$, where it is the unique maximizer: it survives.

Step — knowledge. S^1 uses rationality; S^2 also uses that each firm knows the other is rational; S^k uses $(k-1)$-th order mutual knowledge of rationality. S^∞ needs common knowledge.

Ref: Tadelis (2013) §4.2.1–4.2.2, pp. 63–67; Gibbons (1992) §1.1.B, pp. 4–8.

Grading rubric.

- Recursion stated with the correct crossing ($U_j \to L_i$, $L_j \to U_i$): 0.75 pt.
- Both elimination arguments for all $q_j \in S_j^{k-1}$: 1 pt. Survival of the interior: 0.5 pt.
- Knowledge interpretation: 0.75 pt.
- Recursion only on the symmetric example: max. half the item.

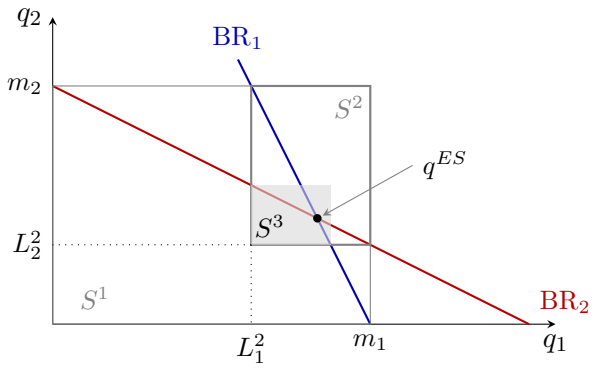
c) (3 points) Characterize S^∞: prove that it is a single profile for every admissible (a, b, c_1, c_2), find it in closed form (including the case in which a firm is driven out), and show that, in the interior case, no finite round leaves a single profile. Draw, in the (q_1, q_2) plane, the two functions BR_1, BR_2 and the sets S^1, S^2, S^3, for $c_1 < c_2$.

Step — contraction. Width $w_i^k \equiv U_i^k - L_i^k = \mathrm{BR}_i(L_j^{k-1}) - \mathrm{BR}_i(U_j^{k-1}) \le \frac{1}{2} w_j^{k-1}$ (slope of BR_i is $-\frac{1}{2}$ or 0). So $w^k \le \max\{m_1, m_2\}/2^{k-1} \to 0$; L_i^k rises and U_i^k falls, so both converge to one point q_i^{ES}, and continuity of BR gives $q_1^{ES} = \mathrm{BR}_1(q_2^{ES})$, $q_2^{ES} = \mathrm{BR}_2(q_1^{ES})$. The composite map has slope at most $\frac{1}{4}$ in absolute value, so this fixed point is unique.

Step — closed form. Interior case, $a - 2c_i + c_j > 0$ for both i: $q_i^{ES} = \dfrac{a - 2c_i + c_j}{3b}$ (substitute one linear branch into the other: $\mathrm{BR}_1(\mathrm{BR}_2(q)) = \frac{a-2c_1+c_2}{4b} + \frac{q}{4}$). If $c_1 \ge (a + c_2)/2$: $(q_1^{ES}, q_2^{ES}) = (0, m_2)$, since $\mathrm{BR}_1(m_2) = \max\{\frac{a-2c_1+c_2}{4b}, 0\} = 0$. Both conditions cannot fail together: their sum is $2a - c_1 - c_2 > 0$.

Step — never finite. Interior case: $L_i^2 = \mathrm{BR}_i(m_j) = \frac{a-2c_i+c_j}{4b} > 0$, so every best reply in the recursion is on its linear branch and $w_i^k = \frac{1}{2} w_j^{k-1}$ exactly: $w_i^k = m_i/2^{k-1}$ (k odd) or $m_j/2^{k-1}$ (k even), positive for every k. The single profile is reached only in the limit. With $c_1 > (a + c_2)/2$, $L_2^k \uparrow m_2 > (a - c_1)/b$, so $U_1^k = 0$ after finitely many rounds: firm 1 is out in finite time.

Step — graph. Drawn with $c_1 < c_2$: $S^1 = [0, m_1] \times [0, m_2]$; each new rectangle uses the previous corners through the best replies.



Ref: Tadelis (2013) §4.2.2, Figure 4.1, p. 67; Gibbons (1992) §1.2.A, Figure 1.2.2, pp. 19–20.

Grading rubric.

- Width contraction and convergence to a fixed point: 1 pt.
- Interior closed form: 0.5 pt; corner case with its condition: 0.5 pt.
- No finite singleton (exact halving): 0.25 pt.
- Graph: axes 0.15, both best replies with correct intercept order 0.25, nested rectangles 0.25, q^{ES} at the crossing 0.1. Corner case omitted: max. 2 pts.

d) (3 points) Now $n \ge 3$ identical firms, each with marginal cost $c \in (0, a)$. Characterize S^∞ and quantify the range of industry output and of the market price compatible with IESDS; compare with $n = 2$.

Step — round 1. The argument of a) holds with $Q_{-i} \equiv \sum_{j \ne i} q_j \ge 0$ in place of q_j: $S_i^1 = [0, m]$, $m = \frac{a-c}{2b}$.

Step — round 2 removes nothing. With every rival in $[0, m]$, Q_{-i} ranges over $[0, (n-1)m] \supseteq [0, 2m] = [0, (a-c)/b]$ because $n - 1 \ge 2$. Each $q \in [0, m]$ is the unique maximizer against $Q_{-i} = (a-c)/b - 2q \in [0, (a-c)/b]$, which rivals can produce (e.g. equal quantities $Q_{-i}/(n-1) \le m$). So $S^2 = S^1$, and $S^\infty = [0, m]^n$.

Step — range. $Q \in [0, \frac{n(a-c)}{2b}]$ and $P \in [\max\{0, a - \frac{n(a-c)}{2}\}, a]$. The set contains zero output ($P = a$), the monopoly output (one firm at m), the competitive output $(a-c)/b$ (two firms at m), and outputs above it ($P < c$). For $n = 2$: $Q^{ES} = \frac{2(a-c)}{3b}$, $P^{ES} = \frac{a+2c}{3}$, a single point.

Economic intuition. With two firms, a bound on the rival's output is a bound on everything the firm faces; with three, two rivals at the top of their range can already flood the market, so no quantity in $[0, m]$ can be ruled out. IESDS exists here but loses uniqueness abruptly at $n = 3$.

Ref: Gibbons (1992) §1.2.A, pp. 20–21; Tadelis (2013) §4.2.3, pp. 67–68.

Grading rubric.

- Round 1 with Q_{-i}: 0.5 pt. Round 2 removes nothing, with the attainability of Q_{-i}: 1.25 pts.
- Output and price ranges with the comparison to $n = 2$: 1.25 pts.
- Duopoly recursion applied to $n \geq 3$ (concluding uniqueness): max. 1 pt.

e) (3 points) Parameter analysis for the interior duopoly. Prove the sign of the effect of a, b, c_1, c_2 on q_1^{ES}, industry output Q^{ES}, the price P^{ES} and firm 1's profit π_1^{ES}. Tabulate, study $c_1 \to c_2$, $c_1 \uparrow (a + c_2)/2$ and the move from $n = 2$ to $n = 3$, and read the results against the uniqueness and invariance criteria.

Step — closed forms. $q_1^{ES} = \frac{a-2c_1+c_2}{3b}$, $Q^{ES} = \frac{2a-c_1-c_2}{3b}$, $P^{ES} = a - bQ^{ES} = \frac{a+c_1+c_2}{3}$, and since $P^{ES} - c_1 = bq_1^{ES}$, $\pi_1^{ES} = b(q_1^{ES})^2 = \frac{(a-2c_1+c_2)^2}{9b}$.

Step — signs. Differentiating: $\partial_a q_1 = \frac{1}{3b}$, $\partial_{c_1} q_1 = -\frac{2}{3b}$, $\partial_{c_2} q_1 = \frac{1}{3b}$, $\partial_b q_1 = -q_1/b$; Q: $+a$, $-c_1$, $-c_2$, $-b$; P: $\frac{1}{3}$ in a, c_1, c_2, and 0 in b; $\pi_1 = bq_1^2$ has the signs of q_1 in a, c_1, c_2 and $\partial_b \pi_1 = -\pi_1/b < 0$.

	a	b	c_1	c_2
q_1^{ES}	$+$	$-$	$-$	$+$
Q^{ES}	$+$	$-$	$-$	$-$
P^{ES}	$+$	0	$+$	$+$
π_1^{ES}	$+$	$-$	$-$	$+$

Step — limits. $c_1 \to c_2 = c$: $q_i^{ES} = \frac{a-c}{3b}$, the symmetric case of the textbooks. $c_1 \uparrow (a + c_2)/2$: $q_1^{ES} \to 0$ and $q_2^{ES} \to m_2$, matching the corner of c) continuously. $n = 2 \to 3$: the prediction jumps from a point to the box $[0, m]^3$ (d).

Step — reading. A rival's cost increase helps a firm; one third of any cost change passes to the price. In costs and demand, the IESDS prediction is invariant (continuous, even through the exit threshold); in the number of firms it fails the uniqueness criterion abruptly. The speed of convergence (width halves each round) does not depend on any parameter.

Grading rubric.

- Closed forms incl. $\pi_1 = bq_1^2$: 0.5 pt. Signs proved: 1.25 pts.
- Table 0.25 pt; limits 0.5 pt; reading against the criteria 0.5 pt.
- Signs without derivation or without reading: max. half the item.

Block 3 — Integrative: spatial competition *(Lecture 1 • 25 points)*

Common setting for Questions 5 and 6. Along a stretch of highway in the interior of São Paulo state, $n \geq 2$ towns are equally spaced (distance 1 between neighbours) and numbered $1, \ldots, n$; unless stated otherwise each town has one unit of households. Two fuel-retail chains, A and B, simultaneously choose one town each in which to open a single station (both may choose the same town). Prices and products are identical and fixed. Every household refuels at the nearest station, measured along the highway, and splits evenly if both are equally distant. Each chain maximizes the number of households it serves. A household's travel cost is its distance to the station it uses; total travel distance is the sum over households.

Question 5. (10 points) Objective items.

a) (3 points) *Normal form × IESDS × welfare.* Suppose instead that the $n \geq 3$ towns are equally spaced around a ring road (like the Rodoanel around São Paulo), with distance measured along the ring. Which statement is correct?

A. A chain serves more households the farther it is from its rival, so IESDS selects antipodal pairs, which also minimize total travel distance.

B. As on a straight highway, IESDS makes the chains cluster in a single town, a profile that maximizes total travel distance.

C. Every profile gives each chain $n/2$ households, IESDS eliminates nothing, and an antipodal pair of stations minimizes total travel distance.

D. Every profile gives each chain $n/2$ households, IESDS eliminates nothing, and total travel distance is the same in every profile.

E. Every profile gives each chain $n/2$ households, so every town is a strictly dominant strategy and every profile is a strict dominant-strategy equilibrium.

Answer: C

Step — payoffs. Same town: $n/2$ each. Otherwise, at ring distance g, the two arcs hold $g-1$ and $n-g-1$ interior towns, each split evenly by its midpoint: $u_A = 1 + \frac{g-1}{2} + \frac{n-g-1}{2} = \frac{n}{2}$.

Step — IESDS. All payoffs are equal, so no strategy is strictly (or weakly) dominated: $S^\infty = S$. Nor is any strategy strictly dominant (E), and no location pattern is selected (A, B).

Step — welfare. An arc with t interior towns between two stations costs $\sum_{s=1}^{t} \min\{s, t+1-s\} = \lfloor (t+1)^2/4 \rfloor$, so gap g costs $f(g) = \lfloor g^2/4 \rfloor + \lfloor (n-g)^2/4 \rfloor$ ($g = 0$: $\lfloor n^2/4 \rfloor$). Using $\lfloor (t+1)^2/4 \rfloor - \lfloor t^2/4 \rfloor = \lfloor (t+1)/2 \rfloor$, $f(g+1) - f(g) = \lfloor \frac{g+1}{2} \rfloor - \lfloor \frac{n-g}{2} \rfloor \leq 0$ for $g \leq \frac{n-1}{2}$: the antipodal gap $\lfloor n/2 \rfloor$ minimizes (D false), clustering maximizes.

Ref: Lecture 1, "Application: spatial competition"; Tadelis (2013) §4.2.3, p. 67.

b) (3 points) *IESDS* × *non-uniform population.* Back on the straight highway, let n be odd and let town 1, a regional capital, have $1 + 2h$ households, with $h \in \{1, \ldots, (n-1)/2\}$; every other town keeps one. IESDS selects:

A. the median town $(n+1)/2$ for both chains, since IESDS depends only on the number of towns

B. town $(n+1)/2 - h$ for both chains

C. town $(n+1)/2 - 2h$ for both chains

D. towns $(n+1)/2 - h$ and $(n+1)/2 - h + 1$ for each chain

E. town 1, the capital, for both chains, for every $h \geq 1$

Answer: B

Step — elimination rule. Let $T = n + 2h$ be total population and $W(\leq \ell)$ the population of towns $1, \ldots, \ell$. In a reduced game with towns $\{\ell, \ldots, r\}$ for both chains, ℓ is strictly dominated by $\ell + 1$ iff $W(\leq \ell) < T/2$: against $s \geq \ell + 2$ moving inward shifts the boundary by half a town and always gains; against $s = \ell + 1$ it compares $W(\leq \ell)$ with $T/2$; against $s = \ell$ it compares $T/2$ with $T - W(\leq \ell)$. Symmetrically on the right.

Step — survivor. IESDS stops at the weighted median: $W(\leq m) = 2h + m \geq T/2 \iff m \geq \frac{n+1}{2} - h - \frac{1}{2}$, and $W(\geq m) = n - m + 1 \geq T/2 \iff m \leq \frac{n+1}{2} - h + \frac{1}{2}$. With n odd, the unique integer is $m^* = (n+1)/2 - h$.

Step — distractors. A ignores the weights; C shifts by the extra households $2h$ instead of half of them; D treats total population as even (two medians); E confuses the largest town with the median.

Ref: Tadelis (2013) §4.2.1, pp. 63–65; Lecture 1, "Application: spatial competition".

c) (2 points, T or F) *Dominance* × *IESDS.* With uniform population and any odd $n \geq 3$, the median town is a strictly dominant strategy for each chain, which is why IESDS selects it.

Answer: F

Step — n = 3. True only here: town 2 earns 2 against a rival at an end and 3/2 against a rival at 2, strictly more than any end town in each case.

Step — n >= 5. Against $s_B = 1$: $u_A(2, 1) = n - 1$, while $u_A(\frac{n+1}{2}, 1) = n - \frac{n+1}{4} < n - 1$ for $n > 3$. The median is not even optimal against every rival location; IESDS selects it only after $(n-1)/2$ rounds, using common knowledge of rationality, not rationality alone.

Ref: Lecture 1, "Iterated elimination of dominated strategies"; Tadelis (2013) §4.1.2, p. 61.

d) (2 points, T or F) *Dominance × policy.* Let $n \geq 4$ be even. A municipal licensing rule allows chain A only in towns $1, \ldots, n/2$ and chain B only in towns $n/2+1, \ldots, n$. Then each chain has a strictly dominant strategy, and the two stations end up in adjacent towns at the centre of the highway.

Answer: T

Step — proof. Now $s_A < s_B$ always, so $u_A = (s_A + s_B - 1)/2$, strictly increasing in s_A for every s_B: $s_A = n/2$ is strictly dominant. $u_B = n - u_A$ is strictly decreasing in s_B: $s_B = n/2 + 1$. Total travel distance is $n(n-2)/4$, against the optimum $\lfloor n^2/8 \rfloor$ of Question 6 d), a ratio that tends to 2: zoning by halves does not undo the clustering.

Ref: Tadelis (2013) §4.1.2, p. 61; Lecture 1, "Application: spatial competition".

Question 6. (15 points) *Integrative: normal form × dominance and IESDS × Pareto optimality × solution-concept criteria (existence, uniqueness, invariance).* Use the common setting above, with uniform population.

a) (3 points) Write the game in normal form for a general n, with each chain's payoff in closed form, and prove that it is constant-sum. Deduce which location profiles are Pareto optimal for the two chains. Who, then, can be harmed by the chains' choices?

Step — normal form. $I = \{A, B\}$, $S_A = S_B = \{1, \ldots, n\}$. If $s_A < s_B$: A serves towns $1, \ldots, s_A$ and half of the $s_B - s_A - 1$ towns strictly between (the midpoint town, if any, is split), so $u_A = s_A + \frac{s_B - s_A - 1}{2} = \frac{s_A + s_B - 1}{2}$. If $s_A = s_B$: $u_A = \frac{n}{2}$. If $s_A > s_B$: $u_A = (n - s_A + 1) + \frac{s_A - s_B - 1}{2} = n - \frac{s_A + s_B - 1}{2}$. u_B follows by symmetry.

Step — constant sum. Every household is served by exactly one chain (or split in halves), so $u_A + u_B = n$ for every profile; the formulas confirm it case by case.

Step — Pareto. If u_A rises, u_B falls by the same amount: no profile Pareto dominates another, so *every* profile is Pareto optimal for the chains. The only parties whose welfare differs across profiles are the households, through travel distance, which no chain internalizes.

Ref: Tadelis (2013) §3.1, pp. 46–49; §3.3.3, p. 56; Lecture 1, "Dominance and social dilemmas".

Grading rubric.

- Three-case payoff with the midpoint split: 1.25 pts (a case missing: −0.5 pt each).
- Constant sum: 0.5 pt. Every profile Pareto optimal for the chains: 0.75 pt.
- Households identified as the harmed party: 0.5 pt.
- "The IESDS profile is Pareto inefficient for the chains": 0 for the Pareto line.

b) (3 points) Characterize S^∞ for every $n \geq 2$: which towns are eliminated in each round, how many rounds there are, and which profiles survive, separately for odd and even n. Prove your characterization. What order of knowledge of rationality does the prediction require?

Step — claim. Round k removes towns k and $n+1-k$ for both chains, as long as at least three towns remain. After k rounds the set is $\{k+1, \ldots, n-k\}$.

Step — extremes dominated. Let the current set be $\{\ell, \ldots, r\}$, $r = n+1-\ell$, $r - \ell \geq 2$ (so $\ell < n/2$). Compare ℓ with $\ell+1$: against $s_B = \ell$, $n/2 < n - \ell$; against $s_B = \ell + 1$, $\ell < n/2$; against $s_B \geq \ell + 2$, $\frac{\ell + s_B - 1}{2} < \frac{\ell + s_B}{2}$. So ℓ is strictly dominated by $\ell + 1$, and r by $r - 1$ by symmetry.

Step — interior survives. A strategy that is optimal against some rival location in the current set cannot be strictly dominated. For $\ell < s < n/2$: against $s_B = s - 1$, locating at s earns $n - s + 1 > n/2$, more than any other town, so s is optimal; symmetrically on the right. The central town(s) are optimal against the rival at the centre: odd n, $(n+1)/2$ earns $n/2 > (n-1)/2$; even n, $n/2$ and $n/2 + 1$ both earn $n/2$ against $n/2 + 1$.

Step — result. Odd n: after $(n-1)/2$ rounds, $S^\infty = \{(\frac{n+1}{2}, \frac{n+1}{2})\}$. Even $n \geq 4$: after $n/2 - 1$ rounds, $S^\infty = \{\frac{n}{2}, \frac{n}{2} + 1\}^2$ (four profiles, each giving $n/2$ to each chain). $n = 2$: every payoff is 1, nothing is removed. Rounds: $\lceil n/2 \rceil - 1$.

Step — knowledge. The last round needs $(\lceil n/2 \rceil - 2)$-th order mutual knowledge of rationality: the longer the highway, the deeper the reasoning the prediction relies on.

Ref: Tadelis (2013) §4.2.1, pp. 63–65; related exercise: Tadelis Exercise 4.10, p. 78.

Grading rubric.

- Extreme dominated by its neighbour, all three cases: 1 pt.
- Interior and central towns survive (optimality argument): 0.75 pt.
- Survivors and number of rounds for both parities and $n = 2$: 0.75 pt. Knowledge order: 0.5 pt.
- Pattern read off small cases without proof: max. a third of the item.

c) (3 points) Replace the towns by a continuum: households are uniformly distributed on $[0, 1]$ (unit mass) and each chain locates anywhere in $[0, 1]$. Characterize S^∞. Compare with b) as the spacing between towns shrinks, and evaluate IESDS against the existence, uniqueness and invariance criteria.

Step — payoffs. $u_A(a, b) = \frac{a+b}{2}$ if $a < b$, $\frac{1}{2}$ if $a = b$, $1 - \frac{a+b}{2}$ if $a > b$.

Step — endpoints removed. 0 is strictly dominated by every $a' \in (0, \frac{2}{3}]$: for $b > a'$, $\frac{a'+b}{2} > \frac{b}{2}$; $b = a'$: $\frac{1}{2} > \frac{a'}{2}$; $0 < b < a'$: $1 - \frac{a'+b}{2} > \frac{b}{2} \iff b < 1 - \frac{a'}{2}$, true for all $b < a'$ iff $a' \leq \frac{2}{3}$; $b = 0$: $1 - \frac{a'}{2} > \frac{1}{2}$. By symmetry 1 is removed: $S_i^1 = (0, 1)$.

Step — nothing else. Take $a \in (0, \frac{1}{2}]$ and any $a' \neq a$. If $a' > a$, against $b = a/2$: $u_A(a', b) = 1 - \frac{a'+b}{2} < 1 - \frac{a+b}{2} = u_A(a, b)$. If $a' < a$, against $b = \frac{a+a'}{2}$: $u_A(a', b) = \frac{a'+b}{2} \leq 1 - \frac{a+b}{2} = u_A(a, b) \iff a + a' \leq 1$, true. Both witnesses lie in $(0, 1)$, so no a' strictly dominates a in any round; $a > \frac{1}{2}$ by symmetry. Hence $S^\infty = (0, 1)^2$.

Step — comparison. With n towns the prediction is the centre after $\lceil n/2 \rceil - 1$ rounds, for every n; in the continuum IESDS removes only the endpoints. The discrete argument relies on a "next town inward" with no rival location strictly between; in the continuum there is always one, where the inward move loses. Existence holds in both; uniqueness is lost in the continuum; and the prediction is not invariant to how finely locations are modelled, since the limit of the discrete predictions is not the prediction of the limit game.

Ref: Tadelis (2013) §3.3.2, pp. 55–56; §4.2.3, pp. 67–68; Cabral (2017) Ch. 14, "Horizontal differentiation", pp. 357–358.

Grading rubric.

- Payoff function: 0.5 pt. Endpoints strictly dominated (with an explicit dominator): 0.75 pt.
- No other strategy dominated, with explicit witnesses for both $a' > a$ and $a' < a$: 1 pt.
- Comparison and the three criteria: 0.75 pt.
- Concluding $S^\infty = \{(\frac{1}{2}, \frac{1}{2})\}$ by analogy with b): max. 1 pt.

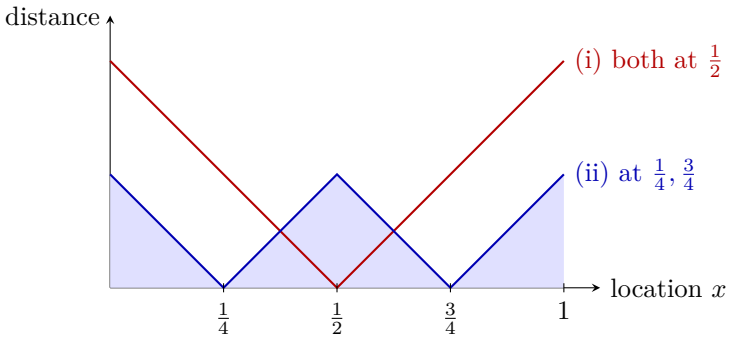
d) (3 points) Welfare. Find the minimum total travel distance $D^{\min}(n)$ over all location pairs and the total travel distance $D^{ES}(n)$ at the IESDS survivors. Prove that $D^{ES}/D^{\min} = 2$ for every odd $n \geq 3$, and characterize the ratio for even n. For the continuum version, draw each household's travel distance as a function of its location when (i) both stations are at $\frac{1}{2}$ and (ii) the stations are at $\frac{1}{4}$ and $\frac{3}{4}$, and read the totals off the figure.

Step — optimum. Given stations $x < y$, households left of the midpoint use x: an optimum splits the towns into a left block of t and a right block of $n - t$, each served from its median. A block of t towns served from its median costs $\lfloor t^2/4 \rfloor$. Minimizing $\lfloor t^2/4 \rfloor + \lfloor (n-t)^2/4 \rfloor$ (same difference argument as Question 5 a) gives $t = \lfloor n/2 \rfloor$ and $D^{\min}(n) = \lfloor n^2/8 \rfloor$.

Step — IESDS. Odd n: both at the median, $D^{ES} = 2\sum_{s=1}^{(n-1)/2} s = \frac{n^2-1}{4}$. Even n: both in one central town, $\frac{n^2}{4}$; in the adjacent central pair, $\frac{n(n-2)}{4}$.

Step — ratio. Odd n: $n^2 \equiv 1 \pmod 8$, so $\lfloor n^2/8 \rfloor = \frac{n^2-1}{8}$ and the ratio is exactly 2. Even n, $4 \mid n$: $D^{\min} = \frac{n^2}{8}$, ratio $\in \{\frac{2(n-2)}{n}, 2\}$. $n \equiv 2 \pmod 4$: $D^{\min} = \frac{n^2-4}{8}$, ratio $\in \{\frac{2n}{n+2}, \frac{2n^2}{n^2-4}\}$, the second above 2 (e.g. $\frac{9}{4}$ at $n = 6$).

Step — continuum figure. Case (i) is the V-shaped $|x - \frac{1}{2}|$, area $\frac{1}{4}$; case (ii) is the W-shaped $\min\{|x - \frac{1}{4}|, |x - \frac{3}{4}|\}$, area $\frac{1}{8}$: ratio 2, as in the odd discrete case.



Economic intuition. Each chain moves inward to steal households from the rival; the households pay for it in kilometres, and the chains, playing a constant-sum game, are indifferent to that cost.

Ref: Lecture 1, "Application: spatial competition"; Cabral (2017) Ch. 14, pp. 357–358.

Grading rubric.

- $D^{\min} = \lfloor n^2/8 \rfloor$ with the block argument: 1 pt.
- D^{ES} for both parities: 0.5 pt. Ratio 2 for odd n proved: 0.5 pt; even case: 0.25 pt.
- Figure: axes 0.25, both curves with kinks at the station sites 0.25, areas read 0.25.
- Optimum guessed at the quartile towns for every n without the block argument: max. 1.5 pts.

e) (3 points) Parameter analysis. Let towns be $d > 0$ apart and hold $N > 0$ households each. Prove how n, d and N affect the number of IESDS rounds K, the IESDS and minimum travel distance per household, and their ratio. Tabulate, study $n = 2$ and $n \to \infty$ (in units of the stretch length $d(n-1)$), and give the economic and policy reading.

Step — scaling. Distances scale with d and the number of households with N; the game's payoffs are multiplied by N, which leaves every dominance relation and so K and S^∞ unchanged: $K = \lceil n/2 \rceil - 1$ depends on n only. Per household (odd n): $\bar{D}^{ES} = d\,\frac{n^2-1}{4n}$ and $\bar{D}^{\min} = d\,\frac{\lfloor n^2/8 \rfloor}{n}$, both increasing in n and in d, neither depending on N; the ratio depends on n only.

	n	d	N
K	+ (weakly)	0	0
$\bar{D}^{ES}$, $\bar{D}^{\min}$	+	+	0
$D^{ES}/D^{\min}$	2 (odd); $\to$ 2 (even)	0	0

Step — limits. $n = 2$: no strategy is removed; the survivors include both the optimum (one station per town, distance 0) and clustering (Nd): IESDS has no bite. $n \to \infty$: relative to the stretch length, $\bar{D}^{ES}/(d(n-1)) \to \frac{1}{4}$ and $\bar{D}^{\min}/(d(n-1)) \to \frac{1}{8}$; both even-$n$ ratios tend to 2.

Step — reading. Clustering at the centre doubles the travel cost, whatever the size of the towns or the distance between them. The chains do not care (constant sum), so the inefficiency is not a social dilemma among the players but an externality on households. A licensing rule by halves keeps the stations adjacent (Question 5 d); restoring the optimum needs the planner to fix both sites. Brazilian law rules out the bluntest instrument, municipal bans on same-sector stores in an area (STF, *Súmula Vinculante* 49: "Ofende o princípio da livre concorrência lei municipal que impede a instalação de estabelecimentos comerciais do mesmo ramo em determinada área").

Grading rubric.

- Invariance of K and S^∞ to N and d argued: 0.75 pt. Per-household signs: 0.75 pt.
- Table 0.25 pt; limits 0.5 pt; reading (players vs households, policy) 0.75 pt.
- Signs without proof or reading: max. half the item.